INF2001
Introduction to Software Engineering



Milestone 2:

Detailed Design,Test Plan and Wireframe

for

<Project title>

Prepared by

Team's Name: ***<replace with your team's name here>***

<name>	**<student ID>**
<name>	**<student ID>**
<name>	**<student ID>**
<name>	**<student ID>**
<name>	**<student ID>**

Lab Group: p*<replace with your lab section>* - *<Team number here>*

Date: *<replace with the date of submission here>*

Table of Contents

Non-Plagiarism Declaration ***3***
AI Usage ***4***
1 Introduction ***5***
1.1 Team member contributions **5**
1.2 Final requirements **5**
1.3 Final Class Diagram **5**
1.4 Component Diagram **5**
2 Detailed Design ***5***
2.1 Architecture Design **5**
2.2 Detailed Design **5**
3 Testing ***6***
3.1 Black Box Testing **6**
3.2 White Box Testing **6**
4 Wireframe ***6***

Non-Plagiarism Declaration

Students are not allowed to copy assessment solutions from another person or other places. It is the students' responsibility to guarantee that their solutions are their own work. Meanwhile, the students must also ensure that their work is not accessible by others. Where such plagiarism is detected, both of the assessments involved will receive ZERO mark.

We hereby declare that:

- We fully understand and agree to the abovementioned policy.
- We did not copy any materials from others or from other places.
- We did not share our materials with others or upload to any other places for public access.
- We agree that we will not disclose any information or material of the team project to others or upload to any other places for public access.
- We agree that our project will receive Zero mark if any misalignment with the abovementioned policies is detected.

Declared by: Group Name (please insert your group name)
Team members:

1. XXX
2. XXX
3. XXX
4. XXX
5. XXX

Date: (please insert the date when you submit your milestone report).

AI Usage

<While it is not prohibited to use AI for the team project, all teams must use AI tools critically and ensure the learning outcomes are achieved.

In case AI tools are used, articulate (i) how and where AI tools are used; and (ii) what changes are made on top of AI outputs. >

1 Introduction

1.1 Team member contributions

<Provide a description of each team member contribution>

1.2 Final requirements

<Provide your project's final requirements. Highlight if there are any updates from M1. Requirements should be written from the systems perspective and meets the 7 traits of good requirements previously covered in the lecture>

1.3 Final Class Diagram

<Provide final updated class diagram. Remember to include <<interface>> classes in your class diagram. Your final class diagram should align with your final requirements and your component diagram>

2 Component Diagram

<Include your component diagram that presents the high-level structure of your system. Provide a short description for the various components and overal design rational. It might be helpful to show the groupings in your class diagram and show the correspondence in the component diagram. Clearly show the interfaces provided and required for each component.>

3 Detailed Design

3.1 Architecture Design

< Describe the overall software architecture of your system. Include an architectural diagram and explain the key architectural components/layers, their responsibilities, and interactions. Provide a brief rationale for selecting this architecture and discuss how it supports your system requirements and use cases. >

3.2 Detailed Design

< Identify and evaluate two design patterns that are applicable to your system. For each pattern, describe its purpose and justify why it is suitable for your design. For the pattern(s) selected for implementation, include the standard pattern template/structure and illustrate how it is applied in your system using relevant class diagram snippets. Focus only on the classes involved in the pattern; the full class diagram is not required. >

4 Testing

4.1 Black Box Testing

<Select ONE of the prominent methods related to your use cases and design the decision table black box testing for it. Show the complete table (causes, effect, combinations). Show the reduced table (if any). Show the final test cases based on the (reduced) table. The number of test cases will depend on the number of final combinations in the table.>

4.2 White Box Testing

<Select ONE of your prominent method (The selected methods for 4.1 and 4.2 should be different. Write the pseudocode for the method. The method should be suitably complex with at least 10 lines of code and 2 decision points. The more the better, but be careful of being too complex! Construct the CFG for your selected method. Calculate the cyclomatic complexity for your method and show the basis path testing and the test cases to cover the paths. >

5 Wireframe

< Provide wireframe screenshots for all implemented use cases. For each wireframe, include a brief description of the screen's purpose, key user interactions, and how it supports the corresponding use case. Ensure all wireframes are consistent with the requirements, use cases, and system design presented in this report. >